

Name: _____

Period: _____

Date: _____

Topic 0.2 Guided Notes: The Scientific Method

Unit 0: Science Practices | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic0.2

Word Bank

<i>Peer Review</i>	<i>Self-Report Bias</i>	<i>Survey</i>	<i>Falsifiable</i>
<i>Generalizability</i>	<i>Social Desirability Bias</i>	<i>Sample</i>	<i>Experimenter Bias</i>
<i>Convenience Sample</i>	<i>Sampling Bias</i>	<i>Operational Definition</i>	<i>Naturalistic Observation</i>
<i>Theory</i>	<i>Hypothesis</i>	<i>Meta-Analysis</i>	<i>Case Study</i>
<i>Replication</i>	<i>Random Sample</i>	<i>Population</i>	<i>Representative Sample</i>

1. Theories, Hypotheses, and Blueprints

A _____ is an explanation using an integrated set of principles that organizes observations and predicts behaviors or events. Because it's too broad to test on its own, we derive a _____ — a specific, testable prediction about the relationship between two or more variables. For it to be scientifically useful, it also needs to be _____ — meaning there must be some possible evidence that could prove it wrong.

To ensure anyone else understands exactly what we mean by a variable, researchers create an _____ — a highly specific, precise description of the exact procedures and measurements used. This allows for _____ — repeating a study to see if the same results can be obtained again. A _____ is a statistical technique that combines and analyzes the results of multiple independent studies. Before publication, a study goes through _____, where other experts evaluate it for accuracy and credibility.

2. Descriptive Research Methods

Fill in the name of each descriptive method next to its description.

- _____: one individual or group is studied in depth in the hope of revealing universal principles — rich detail, but findings may not generalize.
- _____: observing and recording behavior in naturally occurring situations without manipulating the situation. Watch out for _____ — the influence of the experimenter's expectations on the outcome.
- _____: ascertaining the self-reported attitudes or behaviors of a group by questioning a representative sample. Vulnerable to _____ (inaccurate answers) and _____ (answering in a way that makes you look good).

Apply It: Identify the research method

Using the word bank above, write which descriptive method each researcher used.

#	Scenario	Method
1	A psychologist spends six months living with and observing a chimpanzee troop without interfering.	
2	A researcher conducts an in-depth profile of one patient with a rare memory disorder.	
3	A team emails a questionnaire about study habits to 500 randomly selected students.	

3. The Importance of Sampling

Your target _____ is all those in a group being studied. From it, you draw a smaller _____ — a subset of individuals chosen from the larger set. Your goal is a _____, meaning it accurately reflects the demographics of the population as a whole. The best way to achieve this is a _____, where every member of the population has an equal chance of being included.

If researchers get lazy and just test whoever's easy to reach, they're using a _____. This leads to _____ — a flawed process that produces an unrepresentative group. If your sample doesn't look like your population, you lose all _____ — you can't apply your findings to the larger population!

4. AP Science Practice Focus: Identifying Methods and Variables

Practice 2.A asks you to identify a research method or study design — reading a scenario and determining if the researcher used a case study, survey, or naturalistic observation. Practice 2.C asks you to evaluate the appropriate use of research design elements — pulling out operational definitions, potential self-report or social desirability bias, and whether the sampling method allows results to generalize.

The AAQ Exam Connection

On the _____ (AAQ), you'll be given a study summary and asked to dissect it across 6 short free response questions. Part A asks you to identify the _____ used. Part B asks you to pull the _____ directly from the text. Part E asks you to evaluate the study's sampling method to explain its _____.

5. Don't Trip Up! (Common Misconceptions)

Decide which term each description matches, then explain the difference in your own words.

Description	Term
A massive, overarching explanation that organizes a ton of observations (like the Theory of Evolution)	
A tiny, specific, testable prediction derived from a bigger explanation	
The group of people you are actually interacting with in the study	
The massive group of people you are trying to make a general statement about	

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. A team of researchers wants to determine the overall effect of exercise on mood by statistically combining the results of 50 previously published studies on the topic. This process is called a:
 - A. Peer Review
 - B. Meta-Analysis
 - C. Replication
 - D. Case Study
2. A developmental psychologist spends 18 months closely following a single set of conjoined twins, documenting extensive details about their cognitive development, family dynamics, and daily routines. Which research method is being used?
 - A. Survey
 - B. Naturalistic Observation
 - C. Case Study
 - D. Meta-Analysis
3. On an anonymous survey about exercise habits, many participants report exercising far more often than they actually do because they want to appear healthy and disciplined to the researcher. This is best explained by:
 - A. Experimenter Bias
 - B. Social Desirability Bias
 - C. Sampling Bias
 - D. Hindsight Bias
4. A college professor wants to study study habits among all undergraduates nationwide but only surveys the 30 students in his own Monday morning lecture because it is quick and easy. This is an example of a:
 - A. Random Sample
 - B. Convenience Sample
 - C. Meta-Analysis
 - D. Representative Sample
5. Because a study on new medication was conducted using only participants from one small, homogeneous town, the results could not be confidently applied to people in other regions or from different backgrounds. This study is lacking:
 - A. Replication
 - B. Generalizability
 - C. Falsifiability
 - D. Peer Review



Check Your Progress

Done with your notes? Head over to the [Unit 0 Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>